

Motivation:

Lie Groups and Lie Algebras in Physics

Road map

Symmetry means invariance under a transformation. Starting from this physical idea, this opening chapter explains why transformations should be bijections, why bijections naturally form groups, and why physicists need finite groups, discrete groups, Lie groups, Lie algebras, and their representations. The examples include mirror symmetry, Lorentz covariance of the continuity equation, permutations of identical particles, the Monster group, the Standard Model, crystal lattices, the Rubik's cube, matrix representations, particle-hole symmetry, and time reversal. The last section records the course map, expected background, and organisational options appearing in the handwritten lecture.

1.1 Symmetry as invariance under a transformation

Definition 1.1: Symmetry

A *symmetry* of an object or a physical law is a transformation under which the relevant object or law is invariant.

The words “relevant” and “invariant” depend on the problem. A geometric object may be required to occupy exactly the same set of points after a transformation. A physical equation may instead retain the same form in new coordinates, and a quantum state may be unchanged only up to an unobservable phase.

Example 1.1: A mirror-symmetric object

The front elevation of an idealised house is unchanged by reflection in its vertical central

axis.



The transformation moves individual points, but it sends the entire house to the same geometric figure.

Example 1.2: Lorentz covariance of the continuity equation

Let $j^\mu(x)$ be a four-current satisfying

$$\sum_{\mu=0}^3 \partial_\mu j^\mu = 0, \quad \text{or simply} \quad \partial_\mu j^\mu = 0.$$

Under a Lorentz transformation $x'^\mu = L^\mu{}_\nu x^\nu$, a contravariant four-vector and a derivative transform as

$$j'^\mu(x') = L^\mu{}_\nu j^\nu(x), \quad \partial'_\mu = (L^{-1})^\rho{}_\mu \partial_\rho.$$

The transformed divergence is

$$\begin{aligned} \partial'_\mu j'^\mu &= (L^{-1})^\rho{}_\mu \partial_\rho (L^\mu{}_\nu j^\nu) \\ &= (L^{-1})^\rho{}_\mu L^\mu{}_\nu \partial_\rho j^\nu \\ &= \delta^\rho{}_\nu \partial_\rho j^\nu \\ &= \partial_\nu j^\nu = 0. \end{aligned}$$

The Lorentz matrix is constant, so the derivative does not act on it. The middle two matrices multiply to the identity. Hence the continuity equation has the same form in every Lorentz frame. This is invariance of a law rather than invariance of one geometric picture.

Example 1.3: Permutations of identical particles

Consider a system of N particles and a permutation $p \in S_N$. On an ordered list of one-particle data it acts by

$$p \cdot (x_1, \dots, x_N) = (x_{p^{-1}(1)}, \dots, x_{p^{-1}(N)}).$$

If the particles are identical, relabelling them cannot change any physical prediction. In quantum mechanics the permutation operator $P(p)$ acts on the tensor-product Hilbert space. Bosonic states satisfy

$$P(p)\psi = \psi,$$

whereas fermionic states satisfy

$$P(p)\psi = \text{sgn}(p)\psi.$$

Thus permutation symmetry is built into the physical meaning of “identical”.

1.2 What exactly is a transformation?

The most general set-theoretic starting point is a map

$$f : S \longrightarrow S,$$

from a set of objects to itself. Depending on the example, S might be

- a set of geometric objects in Euclidean space;
- a set of four-vector fields $j^\mu(x)$;
- the set of ordered lists describing N particles.

Maps already possess three pieces of the algebraic structure suggested in the handwritten notes.

1. Two maps can be composed:

$$f_1 \circ f_2 : S \longrightarrow S, \quad (f_1 \circ f_2)(x) = f_1(f_2(x)).$$

Composition is closed because the output of f_2 is again an element of S .

2. Composition is associative:

$$(f_1 \circ f_2) \circ f_3 = f_1 \circ (f_2 \circ f_3).$$

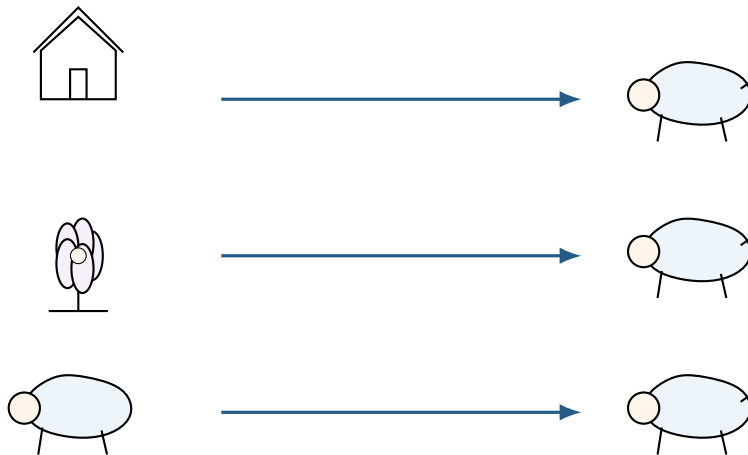
Evaluating either side at x gives $f_1(f_2(f_3(x)))$.

3. There is an identity map

$$\text{id}_S : S \longrightarrow S, \quad \text{id}_S(x) = x,$$

and $f \circ \text{id}_S = \text{id}_S \circ f = f$.

An arbitrary map need not be reversible. The following diagram reproduces the deliberately “bad” actions in the lecture: several distinct objects are sent to the same sheep.



The first two arrows destroy information: from the final sheep one cannot reconstruct whether the initial object was a house, a flower, or a sheep.

图 1.1: Non-bijective maps are not suitable reversible symmetries.

The diagram fails the intuitive requirement of a symmetry because the map is not bijective and therefore has no inverse. We are led to require

$$f : S \longrightarrow S \quad \text{to be bijective.}$$

Every bijection has an inverse bijection, and the inverse undoes the transformation exactly.

Proposition 1.1: Bijections form a group

For every set S , the set

$$\text{Sym}(S) = \{f : S \rightarrow S \mid f \text{ is bijective}\}$$

is a group under composition.

证明. The composition of two bijections is a bijection, so the operation is closed. Function composition is associative. The identity map is bijective and is an identity element. Finally, every bijection f has an inverse bijection f^{-1} , with

$$f \circ f^{-1} = f^{-1} \circ f = \text{id}_S.$$

These are precisely the group axioms. ■

1.3 From transformations to the abstract group concept

The group concept deliberately forgets the internal nature of the objects in S and retains only the transformations and their composition.

Definition 1.2: Group

A *group* is a set G together with a binary operation

$$\circ : G \times G \longrightarrow G, \quad (a, b) \longmapsto a \circ b,$$

such that

1. **associativity:** $a \circ (b \circ c) = (a \circ b) \circ c$ for all $a, b, c \in G$;
2. **identity:** there is $e \in G$ such that $e \circ a = a \circ e = a$ for every $a \in G$;
3. **inverses:** for every $a \in G$, there is $a^{-1} \in G$ such that $a \circ a^{-1} = a^{-1} \circ a = e$.

Closure is encoded by the fact that the operation has codomain G .

The notes place this definition in a short hierarchy:

structure	axioms
magma (called “groupoid” in the lecture)	a set with a closed binary operation
semigroup	a magma with an associative operation
monoid	a semigroup with an identity element
group	a monoid in which every element has an inverse

Modern terminology usually reserves *groupoid* for a category in which every arrow is invertible; *magma* is the unambiguous name for a bare set with a binary operation.

The group axioms are sparse, but their consequences are rich. This is why the same theory applies to permutations, matrices, rotations, lattice symmetries, internal particle symmetries, and many other settings.

1.4 The Monster: how rich can a finite group be?

The lecture pauses at this point to emphasise that an object satisfying only the short group definition can nevertheless have astonishing structure. Its example is the *Monster group*, also called the Fischer–Griess Monster or the Friendly Giant. It is the largest of the 26 sporadic finite simple groups.

Its order is

$$\begin{aligned} |\mathbb{M}| &= 808017424794512875886459904961710757005754368000000000 \\ &= 2^{46} 3^{20} 5^9 7^6 11^2 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \\ &\quad \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71. \end{aligned}$$

This is roughly 8.08×10^{53} . The article reproduced in the notes compares it, evocatively rather than as a precise physical count, to the number of elementary particles in Jupiter.

1.4.1 Several complementary descriptions

The Monster can be approached through several mathematical structures.

1. It is the unique simple group of its order and the largest sporadic simple group.
2. Griess constructed a real algebra of dimension 196884. It is commutative but not associative, and its automorphism group is the Monster. This algebra splits as

$$\mathbb{R}\mathbf{1} \oplus V_{196883},$$

where V_{196883} is the smallest nontrivial irreducible representation highlighted in the lecture.

3. It is the automorphism group of the Monster vertex operator algebra. This is often regarded as the most conceptual construction.
4. The Monster acts by diagram automorphisms of an infinite-dimensional generalised Kac–Moody algebra. This object is called the Monster Lie algebra.

These descriptions illustrate an important theme of the course: an abstract group becomes accessible by allowing it to act on a vector space, algebra, geometry, or physical state space.

1.4.2 The first numerical glimpse of monstrous moonshine

Let

$$q = e^{2\pi i \tau}.$$

The elliptic modular j -function has Fourier expansion

$$j(\tau) = q^{-1} + 744 + 196884q + 21493760q^2 + \cdots.$$

The striking observations are

$$196884 = 196883 + 1$$

and

$$21493760 = 21296876 + 196883 + 1,$$

where the numbers on the right are dimensions of irreducible Monster representations. This suggested a graded representation

$$V = \bigoplus_{n \in \mathbb{Z}} V_n$$

whose graded dimensions reproduce the coefficients of the modular function. The McKay–Thompson series refine the dimension to traces

$$T_g(\tau) = \sum_n \text{tr}(g|_{V_n}) q^n.$$

The eventual construction of the Monster vertex operator algebra and the proof of the moonshine conjectures connected finite group theory, modular forms, vertex algebras, and ideas from string theory.

Why the original article pages are retained

The handwritten lecture embeds a two-page historical article by Richard E. Borcherds, with underlining and annotations. Those pages contain dense typeset material—the exact Monster order, Griess algebra, modular j -function, McKay–Thompson series, vertex algebra, Monster Lie algebra, and references. To preserve every printed and handwritten detail, the two source pages are reproduced below. The surrounding prose gives a beginner-friendly mathematical guide to what is on them.

the concept of a group is extremely rich!

↓ see:

WHAT IS . . .

The Monster?

Richard E. Borcherds

When I was a graduate student, my supervisor John Conway would bring into the department his one-year-old son, who was soon known as the baby monster. A more serious answer to the title question is that the monster is the largest of the (known!) sporadic simple groups. Its name comes from its size: The number of elements is

$$\begin{aligned} &8080, 17424, 79451, 28758, 86459, 90496, \\ &17107, 57005, 75436, 80000, 00000 \\ &= 2^{46} \cdot 3^{20} \cdot 5^9 \cdot 7^6 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot \\ &19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71, \end{aligned}$$

exactly about equal to the number of elementary particles in the planet Jupiter.

The monster was originally predicted to exist by B. Fischer and by R. L. Griess in the early 1970s. Griess constructed it a few years later in an extraordinary tour de force as the group of linear transformations on a vector space of dimension 196883 that preserve a certain commutative but nonassociative bilinear product, now called the Griess product.

Our knowledge of the structure and representations of the monster is now pretty good. The 194 irreducible complex representations were worked out by Fischer, D. Livingstone, and M. P. Thorne (before the monster was even shown to exist). These take up eight large pages in the atlas [A] of finite groups, which is the best single source of information about the monster (and other finite simple groups). The subgroup structure is mostly known; in particular there is an almost complete list of the maximal subgroups, and the main gaps in our knowledge concern embeddings of very small simple groups in the monster. If anyone

wishes to multiply elements of the monster explicitly, R. A. Wilson can supply two matrices that generate the monster. But there is a catch: Each matrix takes up about five gigabytes of storage, and to quote from Wilson's atlas page: "[S]tandard generators have now been made as 196882×196882 matrices over $\text{GF}(2)$ They have been multiplied together, using most of the computing resources of Lehrstuhl D für Mathematik, RWTH Aachen, for about 45 hours...." (The difficulty of multiplying two elements of the monster is caused not so much by its huge size as by the lack of "small" representations; for example, the symmetric group S_{50} is quite a lot bigger than the monster, but it only takes a few minutes to multiply two elements by hand.) Finally the modular representations of the monster for large primes were worked out by G. Hiss and K. Lux; the ones for small primes still seem to be out of reach at the moment.

In the late 1970s John McKay decided to switch from finite group theory to Galois theory. One function that turns up in Galois theory is the elliptic modular function

$$\begin{aligned} j(\tau) &= q^{-1} + 744 + 196884q + 21493760q^2 + \cdots \\ &= \sum c(n)q^n \end{aligned}$$

($q = e^{2\pi i\tau}$), which is essentially the simplest non-constant function invariant under the action $\tau \mapsto (a\tau + b)/(c\tau + d)$ of $SL_2(\mathbb{Z})$ on the upper half plane $\{\tau \mid \Im(\tau) > 0\}$. He noticed that the coefficient 196884 of q^1 was almost equal to the degree 196883 of the smallest complex representation of the monster (up to a small experimental error). The term "moonshine" roughly means weird relations between sporadic groups and modular functions (and anything else) similar to this. It was clear to many people that this was just a meaningless coincidence; after all, if you have enough large integers from various areas of mathematics, then a few are going to be close just by chance, and John McKay was told that his observation was about as useful as looking at tea leaves. John Thompson took McKay's observation further and pointed out that the next few coefficients of the

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¹The announcement of the classification of the finite simple groups about twenty years ago was a little over-enthusiastic, but a recent 1,300-page preprint by M. Aschbacher and S. D. Smith should finally complete it.

图 1.2: The first historical page embedded in the handwritten lecture: Richard E. Borcherds, "What is the Monster?" The lecture's underlining marks the order, the 196883-dimensional representation, the Griess product, and the status of the finite-simple-group classification.

elliptic modular function were also simple linear combinations of dimensions of irreducible representations of the monster; for example, $21493760 = 21296876 + 196883 + 1$. He suggested that there should be a natural infinite-dimensional graded representation $V = \sum_{n \in \mathbb{Z}} V_n$ of the monster such that the dimension of V_n is the coefficient $c(n)$ of q^n in $j(\tau)$, at least for $n \neq 0$. (The constant term of $j(\tau)$ is arbitrary, since adding a constant to j still produces a function invariant under $SL_2(\mathbb{Z})$ and is set equal to 744 mainly for historical reasons.) Conway and Norton [C-N] followed up Thompson's suggestion of looking at the McKay-Thompson series $T_g(\tau) = \sum_n \text{Trace}(g|V_n)q^n$, whose coefficients are given by the traces of elements g of the monster on the representations V_n , and found by calculating the first few terms that these functions all seemed to be *Hauptmoduls* of genus 0. (A *Hauptmodul* is a function similar to j but invariant under some group other than $SL_2(\mathbb{Z})$.) A. O. L. Atkin, P. Fong, and S. D. Smith showed by computer calculation that there was indeed an infinite-dimensional graded representation of the monster whose McKay-Thompson series were the *Hauptmoduls* found by Conway and Norton, and soon afterwards I. B. Frenkel, J. Lepowsky, and A. Meurman explicitly constructed this representation using vertex operators.

If a group acts on a vector space it is natural to ask if it preserves any algebraic structure, such as a bilinear form or product. The monster module constructed by Frenkel-Lepowsky-Meurman has a vertex algebra structure invariant under the action of the monster. Unfortunately there is no easy way to explain what vertex algebras are; see [K] for the best introduction to them. Roughly speaking, vertex algebras can be thought of (at least in characteristic 0) as commutative rings with derivation where the ring multiplication is not quite defined everywhere; this is analogous to rational maps in algebraic geometry, which are also not quite defined everywhere. A more concrete but less intuitive definition of a vertex algebra is that it consists of a space with a countable number of bilinear products satisfying certain rather complicated identities. In the case of the monster vertex algebra $V = \bigoplus V_n$, this gives bilinear maps from $V_i \times V_j$ to V_k for all integers i, j, k , and the special case of the map from $V_2 \times V_2$ to V_2 is (essentially) the Griess product. So the Griess algebra is a sort of section of the monster vertex algebra.

Following an idea of Frenkel, one can use the monster vertex algebra and the Goddard-Thorn "no-ghost theorem" from string theory to construct the *monster Lie algebra*. This is a $\mathbb{Z}^2$ -graded Lie algebra whose piece of degree $(m, n) \in \mathbb{Z}^2$ has dimension $c(mn)$ whenever $(m, n) \neq (0, 0)$. The monster should be thought of as a group of "diagram automorphisms" of this Lie algebra, in the same way

that the symmetric group S_3 is a group of diagram automorphisms of the Lie algebra D_4 . The monster Lie algebra has a denominator formula, similar to the Weyl denominator formula for finite-dimensional Lie algebras and the Macdonald-Kac identities for affine Lie algebras, which looks like

$$j(\sigma) - j(\tau) = p^{-1} \prod_{\substack{m>0 \\ n \in \mathbb{Z}}} (1 - p^m q^n)^{c(mn)}$$

where $p = e^{2\pi i \sigma}$, $q = e^{2\pi i \tau}$. This formula was discovered independently in the 1980s by several people, including M. Koike, S. P. Norton, and D. Zagier. There are similar identities with $j(\tau)$ replaced by the McKay-Thompson series of any element of the monster, and C. J. Cummins and T. Gannon showed that any function satisfying such identities is a *Hauptmodul*. So this provides some sort of explanation of Conway and Norton's observation that the McKay-Thompson series are all *Hauptmoduls*.

So the question "What is the monster?" now has several reasonable answers:

1. It is the largest sporadic simple group or alternatively the unique simple group of its order.
2. It is the automorphism group of the Griess algebra.
3. It is the automorphism group of the monster vertex algebra. (This is probably the best answer.)
4. It is a group of diagram automorphisms of the monster Lie algebra.

Unfortunately none of these definitions is completely satisfactory. At the moment all constructions of the algebraic structures above seem artificial; they are constructed as sums of two or more apparently unrelated spaces, and it takes a lot of effort to define the algebraic structure on the sum of these spaces and to check that the monster acts on the resulting structure. It is still an open problem to find a really simple and natural construction of the monster vertex algebra.

References

- [A] J. H. CONWAY, R. T. CURTIS, S. P. NORTON, R. A. PARKER, and R. A. WILSON, *Atlas of Finite Groups*, Clarendon Press, Oxford, 1985; see also the online Atlas of Finite Group Representations at <http://www.mat.bham.ac.uk/atlas/v2.0/>.
- [C-N] J. H. CONWAY and S. P. NORTON, Monstrous moonshine, *Bull. London. Math. Soc.* **11** (1979), 308–39.
- [K] V. KAC, *Vertex Algebras for Beginners*, second edition, University Lecture Series, vol. 10, American Mathematical Society, Providence, RI, 1998.

The "WHAT IS..." column carries short (one- or two-page), nontechnical articles aimed at graduate students. Each article focuses on a single mathematical object, rather than a whole theory. The *Notices* welcomes feedback and suggestions for topics for future columns. Messages may be sent to notices-what@ams.org.

—Allyn Jackson

图 1.3: The continuation retained from the source lecture. It discusses monstrous moonshine, vertex algebras, the Monster Lie algebra, and four answers to the question "What is the Monster?"

1.5 Why study group theory?

The motivational list in the lecture ranges from the largest sporadic group to familiar laboratory symmetries. Each item points to a later topic.

1.5.1 Finite groups and classification

The Monster motivates the study of finite groups: their examples, properties, subgroups, quotients, representations, and classification. Even when a group is too large to list, its structure can be exposed through smaller subgroups and linear actions.

There are also groups of very different kinds:

$$\mathbb{M}, \quad \text{Sym}(S), \quad SU(N), \quad \dots$$

The first is a particular enormous finite group, the second is the full group of bijections of an arbitrary set, and the third is a continuous matrix group.

1.5.2 The Standard Model and continuous groups

The internal gauge group of the Standard Model is locally described by

$$U(1) \times SU(2) \times SU(3).$$

The three factors encode, respectively, hypercharge, weak isospin, and colour. These are continuous groups: their elements vary smoothly with real parameters. Understanding their local structure leads from *Lie groups* to their infinitesimal *Lie algebras*.

continuous internal symmetries

$$U(1) \quad SU(2) \quad SU(3)$$

one generator three generators eight generators
before symmetry breaking weak sector colour sector

1.5.3 Crystals and discrete groups

Crystallography asks which transformations preserve a lattice in a given dimension. Translations, rotations, reflections, and glide operations form discrete groups because only separated parameter values preserve the lattice.

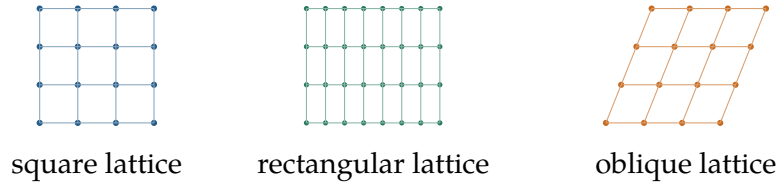


图 1.4: Three of the lattice sketches motivating discrete symmetry groups in crystallography.

1.5.4 The Rubik's cube

Legal moves of a Rubik's cube are permutations of its movable pieces, and sequences of moves form a finite non-Abelian group. The order matters:

$$ab \neq ba$$

for typical face turns a and b . Subgroups describe restricted families of moves, and conjugation

$$aba^{-1}$$

transports a move to a differently oriented position. This physical puzzle therefore gives concrete meaning to noncommutativity, subgroups, and conjugate transformations.

1.6 Representations make abstract groups calculable

Abstract elements need not initially be numbers or matrices. A *representation* replaces each abstract transformation by a linear operator in such a way that composition becomes operator multiplication. This converts group questions into calculations in linear algebra.

The handwritten notes illustrate this by assigning matrices

$$a \hat{=} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad b \hat{=} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

If the representation is D , the rule to preserve is

$$D(a \circ b) = D(a)D(b).$$

In this example,

$$\begin{aligned} D(a)D(b) &= \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}. \end{aligned}$$

The matrix on the last line is the representative of the composite transformation $a \circ b$. Later chapters make this idea precise and study equivalent, reducible, unitary, direct-sum, and tensor-product representations.

1.7 Transformations of quantum systems

The state space of a quantum system supplies many physically important group actions.

Example 1.4: Particle–hole transformation

In fermionic Fock space, a particle–hole transformation exchanges a creation operator with an annihilation operator:

$$c_\alpha \longleftrightarrow c_\alpha^\dagger.$$

Because

$$\{c_\alpha, c_\beta^\dagger\} = \delta_{\alpha\beta}, \quad \{c_\alpha, c_\beta\} = 0,$$

a valid transformation must preserve the canonical anticommutation relations. On a single mode, the number operator changes as

$$n_\alpha = c_\alpha^\dagger c_\alpha \longmapsto c_\alpha c_\alpha^\dagger = 1 - n_\alpha.$$

Thus an occupied state and an empty state are interchanged.

Example 1.5: Time reversal

Time reversal sends

$$t \longmapsto -t.$$

In classical mechanics it also reverses momenta and angular momenta. In quantum mechanics it is represented by an antiunitary operator rather than an ordinary unitary operator. This is an early warning that the mathematical implementation of a symmetry can contain more structure than the visible transformation of coordinates.

These examples motivate representation theory and the systematic study of transformations of quantum systems. The main focus of the lectures is on

Lie groups, and therefore their Lie algebras.

1.8 Map of topics and physical applications

The source lecture organises the motivation in two directions:

1. physical problems identify the mathematics that is necessary;
2. mathematical concepts are repeatedly tested on examples from physics.

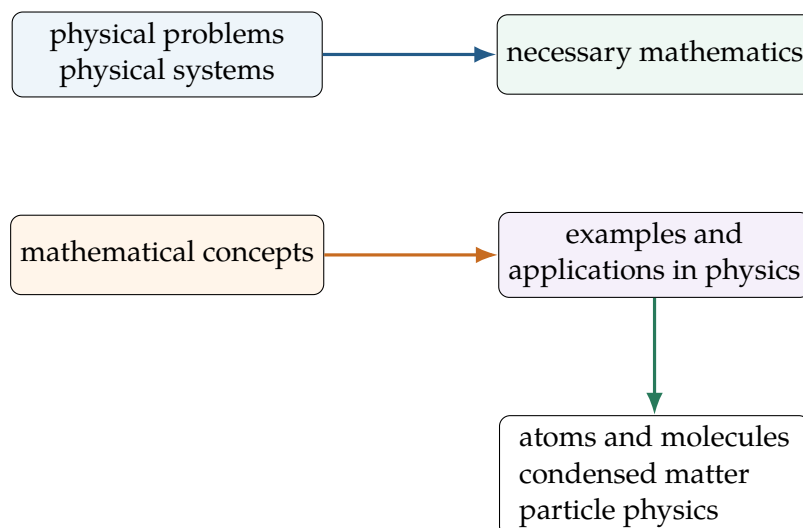


图 1.5: The two-way course philosophy reproduced from the introductory diagram.

The examples on the preceding pages also preview the syllabus:

motivation	topic developed later
Monster group	finite groups
different kinds of groups	examples, properties, and classification
Standard Model gauge group	continuous groups, Lie groups, Lie algebras
crystal lattices	discrete groups
Rubik's cube	basic group concepts
matrix calculations	representation theory
particle-hole and time reversal	transformations of quantum systems

1.8.1 Background checklist from the lecture

The introductory sheet lists the mathematical and physical background that students may bring to the course. The red circles in the source are unfilled checkboxes, so the list records possibilities rather than a claim that every item is mandatory.

Mathematics	Physics
☐ Linear algebra	☐ Theoretical Physics I
☐ Calculus	☐ Theoretical Physics II
☐ Group theory	☐ Theoretical Physics III
☐ Differential geometry	☐ Condensed-matter theory
	☐ Field theory
	☐ Special relativity and four-vector notation
	☐ Second quantisation

The short handwritten phrase “SRT, four notation” has been expanded here as special

relativity and four-vector notation, matching the Lorentz-index calculation at the start of the lecture.

1.9 Course organisation recorded in the source

The last page is an administrative questionnaire. No checkbox is marked, so all alternatives are retained:

Item	Options written in the lecture
Type of examination	☐ oral if the class size $n \leq 10$; ☐ written if $n \geq 10$
Script or printed notes	☐ yes ☐ no
Exercise class and problem solving	☐ at home ☐ in the instructor's presence ☐ no problem-solving session ☐ delegate the problem-solving session
Presentation speed	fast ☐ ☐ slow

A practical reading path

Readers without all the listed background can still use these notes. The early chapters develop abstract group theory from the definitions. Linear algebra is introduced again when matrix groups and representations appear, and the physical sections spell out the needed notation before applying it. The central strategy is always the same: identify transformations, work out how they compose, and then study what remains invariant.

What to retain from this chapter

1. Symmetry is invariance under a transformation. It can mean invariance of a geometric object, covariance of a physical equation, or invariance of physical predictions.
2. Maps $S \rightarrow S$ compose associatively and have an identity map. Requiring transformations to be bijective supplies inverses, so the bijections of S form a group.
3. An abstract group retains the composition law while forgetting what the transformed objects originally were.
4. The Monster demonstrates how much structure can follow from the simple group axioms. Its actions connect finite groups to nonassociative algebras, modular functions, vertex algebras, and Lie algebras.
5. Finite groups, crystallographic groups, and the Rubik's-cube group motivate discrete group theory. The Standard Model group $U(1) \times SU(2) \times SU(3)$ motivates Lie groups and Lie algebras.
6. A representation turns abstract composition into matrix or operator multiplication:

$$D(ab) = D(a)D(b).$$

7. Particle-hole symmetry and time reversal illustrate why quantum mechanics needs a careful theory of group actions on state spaces.